

JOMFRULAND FYR, Norway

WMO: 014760

Lat: 58.857N

Lon: 9.576E

Elev: 5

StdP: 101.26

Time Zone: 1.00 (EUC)

Period: 02-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-10.0	-7.9	-13.7	1.1	-7.4	-11.5	1.4	-4.7	13.8	3.7	12.5	3.4	4.4	30	0.666

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	5.2	23.6	17.2	22.2	17.1	20.9	16.3	19.2	22.1	18.3	20.9	17.5	20.0	3.7	80

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.0	13.0	20.9	17.2	12.3	20.1	16.3	11.6	19.1	54.7	22.2	51.7	20.9	49.2	20.0	21.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	12.1	10.8	-10.1	26.1	2.6	1.7	-12.0	27.3	-13.6	28.3	-15.0	29.2	-16.9	30.4		
(5)			-10.8	19.5	2.6	1.2	-12.7	20.4	-14.3	21.1	-15.8	21.8	-17.7	22.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	8.3	0.3	-0.1	2.4	6.6	11.1	15.2	17.4	17.0	13.9	8.8	4.8	1.7	(6)
(7)		DBStd	7.03	4.15	3.80	3.45	2.70	3.09	2.27	2.21	1.87	2.16	2.99	3.48	4.54	(7)
(8)		HDD10.0	1425	300	284	236	108	21	0	0	0	1	59	158	258	(8)
(9)		HDD18.3	3694	558	517	493	353	225	99	46	50	134	297	407	516	(9)
(10)		CDD10.0	802	0	0	1	5	55	156	229	218	117	21	1	0	(10)
(11)		CDD18.3	32	0	0	0	0	1	4	17	9	0	0	0	0	(11)
(12)		CDH23.3	59	0	0	0	0	3	9	42	4	0	0	0	0	(12)
(13)		CDH26.7	2	0	0	0	0	0	0	2	0	0	0	0	0	(13)
(14)	Wind	WSAvg	4.9	5.3	4.7	4.6	4.2	4.5	4.4	4.7	5.2	5.3	5.7	5.3	(14)	
(15)	Precipitation	PrecAvg	1115	106	68	46	61	74	78	89	121	111	133	132	93	(15)
(16)		PrecMax	1588	266	224	122	177	200	182	195	222	310	347	490	169	(16)
(17)		PrecMin	783	10	6	0	4	6	7	14	19	43	18	30	29	(17)
(18)		PrecStd	209	67	48	36	41	43	41	47	58	68	74	93	43	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	9.2	13.3	16.3	23.0	24.7	26.5	23.8	20.4	15.6	12.8	10.6	(19)
(20)			MCWB	6.9	5.4	7.7	10.2	15.7	16.5	18.7	15.2	14.2	10.8	10.8	8.8	(20)
(21)		2%	DB	7.6	6.9	9.8	13.9	19.9	22.4	24.5	22.5	18.9	14.6	11.6	9.4	(21)
(22)			MCWB	5.8	4.3	5.6	8.8	14.6	16.6	18.6	16.5	14.1	11.6	10.1	7.9	(22)
(23)		5%	DB	6.8	5.6	7.9	12.2	17.7	20.5	22.6	21.4	17.9	13.8	10.5	8.4	(23)
(24)			MCWB	5.2	4.1	4.3	8.1	13.3	15.3	18.1	16.5	13.9	11.3	9.1	7.0	(24)
(25)		10%	DB	5.8	4.6	6.4	10.8	15.9	19.0	21.2	20.2	17.1	13.1	9.6	7.5	(25)
(26)			MCWB	4.3	3.4	4.0	7.3	11.9	14.3	17.0	16.0	13.7	10.6	8.3	6.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.5	6.1	8.2	11.5	17.3	19.0	20.6	19.7	17.1	14.2	11.6	9.4	(27)
(28)			MCDWB	8.5	7.4	12.6	14.5	21.7	22.6	24.1	22.0	18.6	14.6	12.3	10.2	(28)
(29)		2%	WB	6.5	5.1	6.3	10.0	15.6	17.3	19.7	18.8	16.2	13.1	10.4	8.3	(29)
(30)			MCDWB	7.5	6.3	8.6	12.9	19.2	20.7	23.1	21.0	17.6	13.9	11.3	9.2	(30)
(31)		5%	WB	5.6	4.4	5.3	8.9	14.0	16.2	18.8	18.1	15.5	12.3	9.5	7.4	(31)
(32)			MCDWB	6.5	5.5	7.0	11.4	16.9	19.5	21.8	20.2	17.0	13.3	10.5	8.3	(32)
(33)		10%	WB	4.5	3.5	4.6	7.9	12.7	15.3	17.8	17.3	14.7	11.4	8.5	6.3	(33)
(34)			MCDWB	5.6	4.6	6.0	10.2	15.3	18.1	20.6	19.4	16.5	12.7	9.6	7.4	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	3.8	3.9	5.0	5.4	5.5	5.6	5.2	5.0	4.7	4.2	3.7	3.9	(35)
(36)			MCDBR	4.3	5.2	7.5	8.0	8.2	7.8	7.4	6.1	5.5	4.7	4.5	4.0	(36)
(37)			MCWBR	4.2	3.7	4.6	4.5	4.5	4.4	3.8	3.5	3.6	3.7	4.2	4.0	(37)
(38)		5% WB	MCDWR	4.0	4.5	6.3	7.0	7.6	6.8	6.3	5.1	4.6	4.0	4.1	3.8	(38)
(39)			MCWBR	4.2	3.5	4.2	4.4	4.5	4.3	3.6	3.6	3.8	3.9	4.2	4.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.320	0.340	0.379	0.372	0.357	0.378	0.377	0.349	0.336	0.302	0.263	(40)	
(41)		taud	2.240	2.321	2.369	2.325	2.415	2.478	2.438	2.438	2.506	2.444	2.299	2.175	(41)	
(42)		Ebn at Noon	532	700	802	825	860	879	848	821	791	681	509	424	(42)	
(43)		Edh at Noon	45	68	86	106	104	99	102	95	76	61	42	33	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.33	0.93	2.40	3.93	5.12	5.86	5.27	4.09	2.65	1.22	0.44	0.23	(44)	
(45)		RadStd	0.07	0.20	0.21	0.55	0.59	0.43	0.57	0.27	0.33	0.21	0.11	0.03	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page